



RISE IN × STELLAR PRO HACKATHON · 19–20 SEP 2026

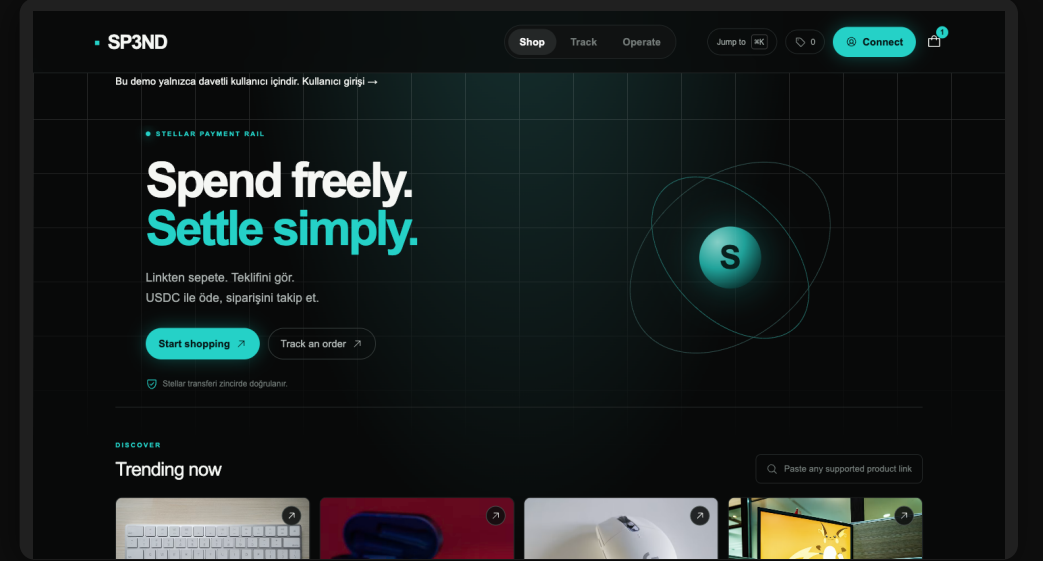
Spend freely. Settle simply.

One USDC payment on Stellar. Real goods from Amazon Türkiye, Trendyol, and Hepsiburada — delivered to a Turkish address.

LOCAL COMMERCE

STELLAR USDC

FREIGHTER + SEP-53



Working app in-repo · github.com/HDemir23/SP3ND · sp3nd.shop

USDC sits in wallets. Shopping still needs a card.

THE LOCK

Amazon Türkiye, Trendyol, and Hepsiburada do not take Stellar USDC. A holder who wants a real product still has to off-ramp, wire TRY, and check out as a card customer.

THE COST

Each hop adds FX spread, banking hours, and KYC. Remittances become cash, not the actual goods a family asked for. Crypto-native users bounce before they spend.

THE GAP

Stellar already moves dollars in seconds. Turkey already has dense e-commerce. Nobody connects the two without asking the shopper for a card or a private key.

Use case: local payments + commerce. Not a demo swap. A rail that turns Stellar dollars into a parcel at a Turkish door.

A clear buyer — and a desk that can serve them this week

01 USDC holders in Turkey

People who already keep dollars on Stellar and want a keyboard, a phone case, or a gift without opening a bank transfer.

02 Diaspora sending goods, not cash

A relative abroad pays in USDC. The parcel lands at a Turkish address. Remittance becomes a delivered product.

03 Crypto-native shoppers

Paste a product URL, review a 15-minute quote, sign once in Freighter. No card number ever enters the app.

04 The operator desk

A human merchant-of-record who already has store accounts. They buy, attach real order refs, and ship. The protocol does not scrape cards.

Paste a link. Pay USDC once. Get the thing.

3 stores

One cart

Amazon TR, Trendyol,
Hepsiburada in a single quote.

1 transfer

One USDC total

Goods + shipping + 1 USDC
per unit. No second charge.

0 cards

Keys stay local

Freighter signs. The app
never sees a secret or a PAN.

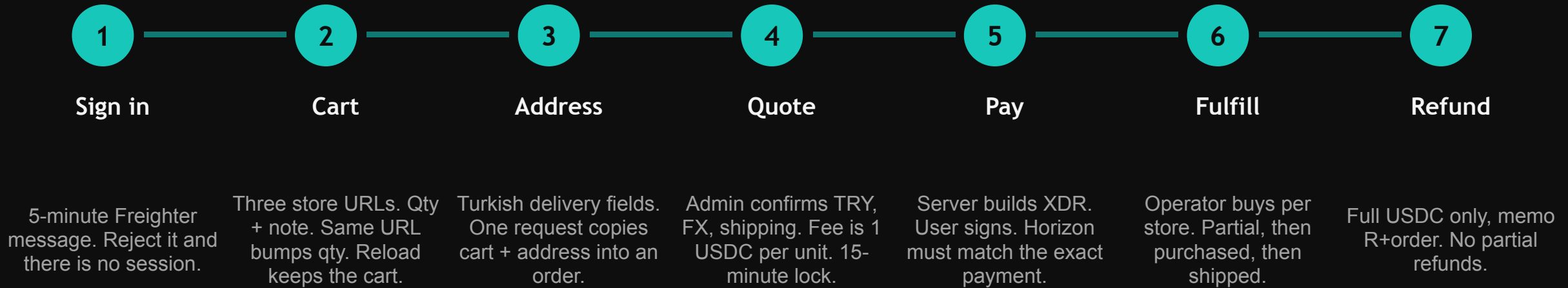
15 min

Firm quote

Admin-verified TRY prices,
TCMB FX, then a locked
USDC amount.

What we refused to fake: auto-buy on merchant sites, escrow we do not yet have, and partial refunds. The operator buys with their own account. The chain only moves USDC.

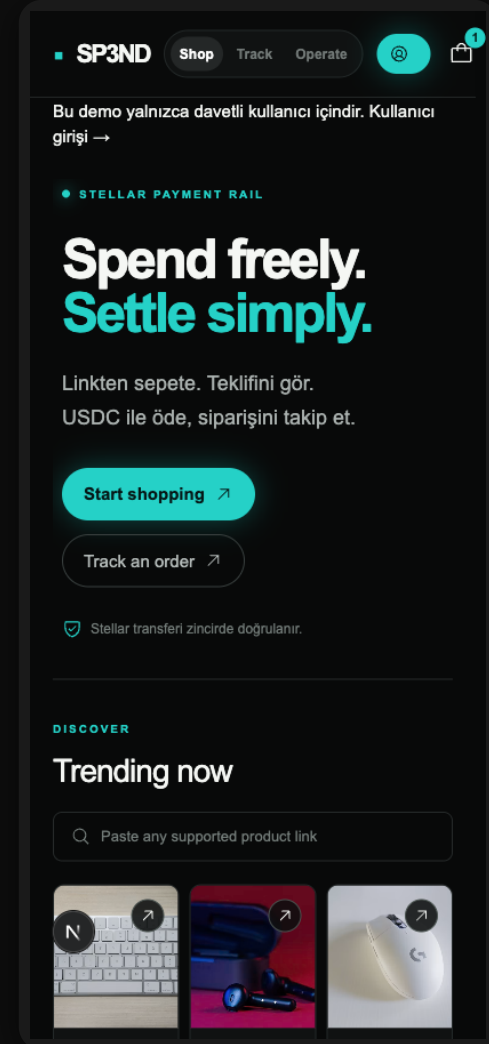
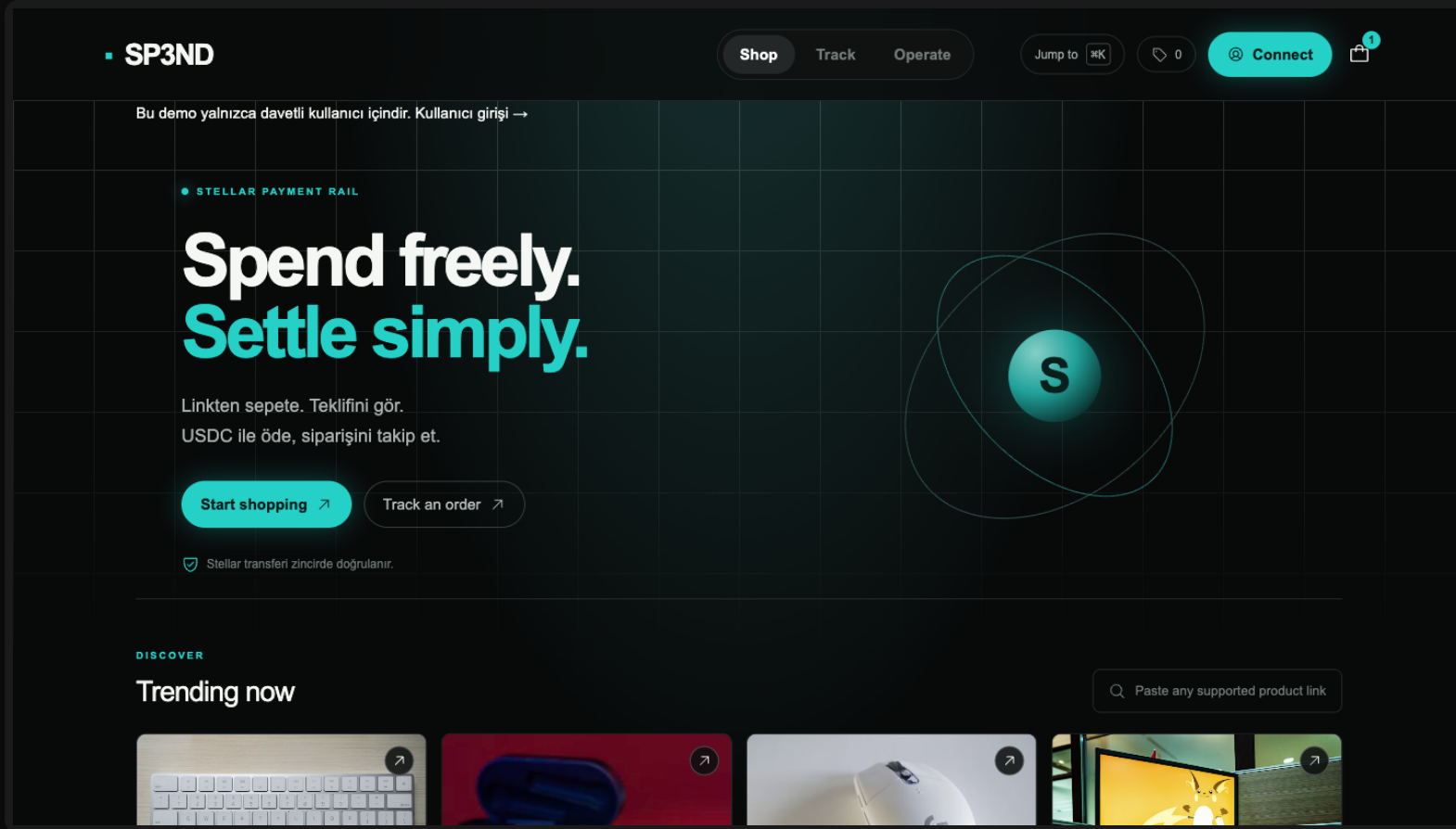
Seven steps judges can click without a script rewrite



Wallet isolation is part of the product, not a footnote.

A second Freighter account must not see the first wallet's cart, address, or orders. Switching accounts forces a new sign-in. Payment XDR is bound to the session wallet. Chrome extension only has activeTab — it never stores a session.

A shop a non-crypto relative can finish



Mosaic storefront · paste-a-link cart · Chrome extension · Turkish copy throughout

The quote is a formula. The payment is a real USDC operation.

SERVER-SIDE TOTAL

$$\text{USDC} = \frac{(\text{goods TRY} + \text{shipping TRY})}{\text{USDTRY}} + 1 \text{ USDC} \times \text{quantity}$$

Service fee	1 USDC per unit. Client cannot override it.
FX	Admin confirms TCMB USD selling rate (4 dp).
Rounding	Decimal.js, USDC rounded up to 7 dp.
Lock	15 minutes. Timebounds on the Stellar tx.
XDR	Built on the server. Freightier only signs.
Paid status	Only after Horizon matches hash, payer, receiver, issuer, amount, memo, time.

No double pay

Network delay keeps watching the same hash. A second payment is not prepared.

Refresh-safe

Reload continues polling. Quote/cancel stay locked while an attempt is live.

Reject-safe

If Freightier refuses, the same XDR can be signed again.

Unlock rule

After expiry + 60s of Horizon ledger time, the hash is checked, then the lock lifts.

Classic Stellar payments, a Next.js desk, no secrets on the server

CLIENT

Next.js 16 · Freighter API
Chrome extension (activeTab)

APP

Route handlers · Prisma
Wallet-scoped sessions (8h)

DATA

SQLite · cart, orders, memos
Login challenges, payment attempts

STELLAR

stellar-sdk · Horizon
USDC payment + memo + timebounds

AUTH

SEP-53 message signing
SHA-256 + Ed25519, one-time nonce

FX / CATALOG

TCMB USD selling rate
Amazon TR fetch · Rainforest fallback

OPERATOR

Admin quotes, store-by-store purchase refs, shipping, full refund
(R + memo)

MERCHANTS

Amazon Türkiye · Trendyol · Hepsiburada — bought with the
operator's own account, never the app's

Stellar is the checkout, not a badge on the footer

Freighter

Wallet connect, message sign-in, and the USDC payment signature. Nothing is custodied.

SEP-53

Documented signed-message format. Origin + wallet + nonce + 5-minute expiry. Atomic one-time use.

Circle USDC

Testnet issuer GBBD47...FLA5 · Mainnet issuer GA5ZSEJY...KZVN. Trustlines checked on both sides.

Horizon + SDK

@stellar/stellar-sdk builds the XDR, submits, and verifies the exact payment operation.

Memo + bounds

Order memo on the payment. maxTime = quote expiry. Replay and overpay are rejected.

Skills / docs

Stellar signing docs, SEP-53, Freighter API, Horizon. TR Mock Anchor is next — not a mocked SEP-6 today.

Not mocked. Not hardcoded. Deliberately scoped.

Real Horizon path

npm run test:testnet funds Friendbot accounts, opens trustlines, issues demo USDC, and drives the HTTP API against live testnet.

What is out of scope

No Soroban contract in this MVP. No SEP-6 deposit. No automatic merchant purchase. No partial refund. No card vault.

Unit + API tests

npm test uses temp SQLite and a fake Horizon. npm run test:api signs two wallets and asserts isolation, then deletes its rows.

Why that is a feature

A broken escrow or a scraped store login would fail the demo. A signed USDC payment and a human desk will not.

Idempotent checkout

Wallet-bound checkout key returns the same order on retry. Cart lines unique on wallet + URL + notes.

Passkeys / smart wallets

Treated as a bonus, as the brief asks. Freightier is the supported signer today.

Ship the desk, then put TRY on-ramp and escrow on-chain

NOW

Freighter login + USDC pay

3-store cart, TR address

Admin quote / buy / ship / refund

Chrome extension

NEXT 90 DAYS

TR Mock Anchor (SEP-6 / 10 / 38)

TRY deposit → USDC in-app

Soroban escrow: release on fulfill

Live small mainnet orders

SCF / INSTAAWARDS

Licensed TRY anchor partner

More TR merchants

Diaspora GTM in EU/UK/DE

Passkey smart wallets

Intended next step: Stellar Community Fund, with InstaAwards as the fast path if the live desk proves a handful of real parcels.

The acceptance script — same order as DEMO.md

- 1 Sign in** Chrome + Freighter. Sign the 5-minute message. Reject it once so judges see no session.
- 2 Cart** Amazon TR, Hepsiburada, Trendyol — one each. Qty + color/size note. Repeat URL bumps qty. Reload. Extension fetch + confirm.
- 3 Address** Fill a Turkish address. One quote request. Order shows three merchants + the address.
- 4 Quote** Admin: name, unit TRY, USDTRY, per-store shipping. Fee = 1 USDC/unit. User sees breakdown + 15:00.
- 5 Pay** Freighter: wallet, network, receiver, amount. Sign a small real payment. Explorer link. Same hash if delayed.
- 6 Fulfill** Operator buys on store accounts. Attach real refs. Partial then purchased, then partial/full shipped.
- 7 Refund** If needed: full refund only. Send R+memo from the desk wallet. Panel verifies the hash.

Everything a judge needs to run, click, and score

GitHub `github.com/HDemir23/SP3ND`

Public site `www.sp3nd.shop`

Local app `localhost:3000` after `npm ci && npm run setup && npm run dev`

Admin `/login?role=admin` · password from `.env`

Extension `/extension` · or `chrome://extensions` load unpacked

Verify `npm test` · `npm run typecheck` · `npm run test:api` · `npm run test:testnet`



Spend freely. Settle simply.

Stellar already moves the dollar. SP3ND spends it at a Turkish door.

Repo

github.com/HDemir23/SP3ND

Site

sp3nd.shop

Next

[SCF](#) · [Anchor](#) · [escrow](#)

[Thank you](#) · [questions](#)